

Aayush Patel

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ML engineer at Georgia Tech — LLM fine-tuning, RL systems, production RAG pipelines, end-to-end feature engineering

EDUCATION

Georgia Institute of Technology — Atlanta, GA

Expected May 2028

B.S. Computer Science — Intelligence & Systems Architecture / Minor: FinTech / GPA: 3.71 / 4.00

Coursework: Machine Learning, Artificial Intelligence, Advanced Data Structures & Algorithms, Objects & Design

TECHNICAL SKILLS

ML & AI Systems: PyTorch, scikit-learn, Pandas, NumPy, Retrieval-Augmented Generation (RAG), Prompt Engineering, Structured Extraction, LLM API Integration (OpenAI, Gemini, Claude, Vertex AI), Vector Search (pgvector/HNSW, ChromaDB), LSTM, ARIMA, Random Forest, XGBoost, CNN, Evolutionary Algorithms (deap), NLP, QLoRA/SFT/DPO Fine-Tuning, Reinforcement Learning (PPO/SAC, Gymnasium)

Languages: Python, SQL, Java, C, C++, JavaScript, TypeScript

Infrastructure & Tools: FastAPI, Flask, PostgreSQL, pgvector, Docker, Docker Compose, AWS, CI/CD (GitHub Actions), Google Cloud Platform (GCP), REST APIs, Next.js, Firebase/Firestore, Pydantic, WebSockets, Git

WORK EXPERIENCE

Georgia Institute of Technology

May 2026 – Aug 2026

Teaching Assistant, CS 2340: Objects & Design — Atlanta, GA

- Developed OOP, design-pattern, and Agile/Scrum materials for 200+ students across CS 2340 sections
- Recognized by faculty for documenting software design trade-offs to make it clear to students of varied backgrounds

Big Data Big Impact, Georgia Tech

Jan 2026 – Present

Director of Projects — Atlanta, GA

- Led 11 AI/ML project teams (100+ students) across NLP, forecasting, and computer-vision domains
- Set uniform model-evaluation standards: precision/recall dashboards, confusion matrices, stratified CV, AUC score
- Drove 6 projects to deployment; presented results to College of Computing Advisory board and a Microsoft industry panel

Georgia Tech VIP — EMADE Lab (Advisor: Dr. Jason Zutty)

Aug 2025 – May 2026

Undergraduate Researcher — Evolutionary AutoML & LLM-Guided Synthesis — Atlanta, GA

- Designed LLM-as-mutation-operator framework; integrated Gemma, Qwen, Gemini, and DeepSeek-R1 in *deap* framework
- Improved synthesis solution quality by 14% on TSP and symbolic regression over human-engineered baselines
- Engineered island-migration scheduler for parallel evolution; achieved 5.8× convergence speedup

Mascot Forge Pvt Ltd (automotive-parts manufacturer, India)

May 2025 – Aug 2025

Data Engineering Intern

- Built gradient-boosted demand-forecasting model with scikit-learn on production records requiring rigorous pre-processing
- Validated on held-out data; model adopted by operations management, cutting reporting latency by 1.3 hours
- Built PostgreSQL analytics dashboards; translated model findings into executive-level ROI presentations

PROJECTS

FinLoRA — SEC 10-K Financial QA Fine-Tuning *Python / QLoRA/DPO / Phi-3.5-mini / Hugging Face / FastAPI / vLLM*

- Fine-tuned Phi-3.5-mini (3.8B) via QLoRA SFT+DPO on 34,799 SEC/FinQA examples; deployed live via FastAPI+vLLM
- Achieved +88% ROUGE-1 and +8.7pp BERTScore over base; built EDGAR pipeline with MinHash LSH deduplication

RLPortfolio — Regime-Aware RL Trading Agent *Python / PPO/SAC / Gymnasium / HMM / FastAPI / Stable-Baselines3*

- HMM-conditioned PPO agent over 10 ETFs; Sharpe 0.508, +12.2% annualized factor alpha (p = 0.013) on 7-year OOS eval
- Validated via 15 expanding walk-forward windows with circular block bootstrap CIs and HAC-robust factor regression
- Composite reward (Sharpe + drawdown + turnover penalties) with per-ETF microstructure costs; served via FastAPI

Research Paper RAG Platform

FastAPI / pgvector / Next.js / Docker / Claude API

- Production RAG pipeline: page-aware PDF chunking, pgvector/HNSW index; achieved Recall@5 of 0.82 and MRR 0.76
- Weak-evidence flagging and field-level citations via Pydantic-validated extraction; containerized full stack with Docker

TerraTrends — County-Level Economic Forecasting

Python / LSTM/ARIMA / scikit-learn / Flask / deck.gl / GCP

- Trained LSTM/ARIMA ensemble on multi-source economic features; automated ETL from 15+ public datasets on GCP

Flight Delay Prediction

Python / scikit-learn / XGBoost / Flask / Pandas

- Trained CNN, RF, XGBoost, and MLP ensemble on 6M+ flight-weather records; achieved 92% accuracy and AUC 0.94